

HOEK VAN HOLLAND, Netherlands

WMO: 063300

Lat: 51.9911N

Lon: 4.1217E

Elev: 14

StdP: 101.16

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.8	(c) -3.1	(d) -9.8	(e) 1.6	(f) -3.0	(g) -7.1	(h) 2.1	(i) -2.1	(j) 18.7	(k) 6.4	(l) 17.3	(m) 6.6	(n) 5.5	(o) 80	(p) 0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.6	(c) 27.5	(d) 19.4	(e) 25.0	(f) 18.5	(g) 23.0	(h) 18.0	(i) 20.6	(j) 25.5	(k) 19.6	(l) 23.5	(m) 18.7	(n) 21.9	(o) 5.2	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	22.4	18.2	13.1	21.4	17.4	12.5	20.5	59.5	25.6	56.0	23.7	53.2	22.0	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	14.3	13.0	-6.7	33.4	2.5	2.0	-8.5	34.8	-10.0	36.0	-11.4	37.1	-13.2	38.5
			-7.3	22.8	2.6	0.9	-9.2	23.5	-10.7	24.0	-12.1	24.6	-14.0	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	4.3	4.8	6.9	10.1	13.2	15.8	18.1	18.4	16.0	12.3	8.3	5.2
	DBStd	5.87	3.61	3.34	2.97	3.29	3.22	2.69	2.61	2.35	2.38	2.94	3.03	3.67
	HDD10.0	699	177	147	103	38	6	0	0	0	0	12	65	151
	HDD18.3	2704	434	380	356	249	163	87	34	27	78	188	300	407
	CDD10.0	1118	1	1	6	40	105	173	252	261	179	83	15	2
	CDD18.3	80	0	0	0	1	4	10	28	29	7	1	0	0
	CDH23.3	419	0	0	0	4	36	65	150	137	27	1	0	0
	CDH26.7	114	0	0	0	0	6	18	47	38	6	0	0	0
Wind	WSAvg	6.8	7.8	7.4	7.0	6.3	6.5	6.3	6.0	6.0	6.4	7.1	7.3	7.7
Precipitation	PrecAvg	879	73	65	49	40	56	70	82	97	86	83	96	84
	PrecMax	1184	168	123	102	92	106	136	174	204	243	179	159	161
	PrecMin	619	7	19	11	0	19	23	17	22	18	13	7	15
	PrecStd	126	37	27	25	23	25	35	42	55	56	40	37	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	13.5	18.0	23.5	27.0	29.3	31.5	30.7	26.4	21.7	16.0	12.9
		MCWB	10.3	9.2	11.5	14.2	16.5	20.1	20.6	20.5	19.4	17.3	13.8	11.3
	2%	DB	10.4	11.1	14.4	20.0	23.5	24.8	27.0	26.9	23.0	18.8	14.2	11.7
		MCWB	9.3	8.8	10.3	13.1	16.1	18.4	19.5	19.7	18.0	15.7	12.7	10.4
	5%	DB	9.5	9.8	12.2	17.4	20.5	22.0	24.3	24.1	21.2	17.3	13.3	10.7
		MCWB	8.4	8.3	9.3	12.0	14.9	17.2	19.1	19.0	17.3	15.0	11.8	9.6
	10%	DB	8.5	8.9	10.7	14.9	18.1	20.0	22.3	22.2	19.6	16.1	12.3	9.7
		MCWB	7.5	7.5	8.6	11.1	14.1	16.4	18.2	18.0	16.4	14.2	10.8	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	10.2	12.6	15.3	18.8	21.1	22.0	21.9	20.3	17.6	14.2	11.7
		MCDB	11.3	11.8	15.9	21.3	25.0	27.1	27.5	27.7	25.1	20.5	15.2	12.5
	2%	WB	9.5	9.3	10.9	13.6	16.7	19.2	20.7	20.6	18.9	16.3	13.0	10.7
		MCDB	10.2	10.6	13.3	18.6	21.1	23.7	25.5	25.0	22.1	18.2	14.0	11.5
	5%	WB	8.5	8.5	9.8	12.5	15.5	17.7	19.6	19.6	17.9	15.4	12.1	9.8
		MCDB	9.2	9.6	11.7	16.4	19.4	21.1	23.3	23.0	20.4	17.0	13.1	10.6
	10%	WB	7.7	7.7	8.8	11.5	14.4	16.7	18.6	18.7	17.0	14.6	11.1	8.7
		MCDB	8.5	8.8	10.4	14.3	17.8	19.3	21.6	21.5	19.2	16.0	12.1	9.6
Mean Daily Temperature Range	5% DB	MDBR	3.7	4.4	5.5	6.9	6.8	6.4	6.3	6.6	6.1	5.3	4.1	3.7
		MCDBR	3.7	5.3	8.2	11.2	11.2	11.0	10.6	10.6	8.7	6.3	4.4	3.5
		MCWBR	3.5	3.8	5.0	5.5	5.4	5.8	5.1	5.1	4.7	4.1	3.7	3.4
	5% WB	MCDBR	3.6	4.6	7.0	9.7	9.8	9.3	8.9	8.8	7.6	5.9	4.1	3.5
		MCWBR	3.6	3.8	4.7	5.1	5.1	5.4	4.8	4.8	4.5	4.2	3.8	3.6
Clear-Sky Solar Irradiance	taub	0.339	0.358	0.391	0.415	0.423	0.418	0.425	0.416	0.395	0.373	0.350	0.334	
	taud	2.338	2.333	2.256	2.228	2.269	2.299	2.288	2.327	2.376	2.393	2.356	2.326	
	Ebn at Noon	649	748	792	818	832	838	824	811	784	722	628	584	
	Edh at Noon	64	84	109	126	127	124	124	113	96	79	62	55	
All-Sky Solar Radiation	RadAvg	0.71	1.41	2.64	4.30	5.18	5.55	5.25	4.44	3.21	1.82	0.83	0.54	
	RadStd	0.07	0.26	0.32	0.41	0.36	0.38	0.56	0.35	0.25	0.16	0.08	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.81	N/A	N/A	N/A	N/A	-113	+66	N/A
(47) Regional (80 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page